

2 Marks answers

1. What is AI?

Definition:

Artificial Intelligence (AI) is the branch of computer science that enables machines to perform tasks that normally require human intelligence.

Explanation:

It includes learning, reasoning, problem-solving, and decision-making.

👉 Example: Chatbots, Google Maps.

2. Why do we need to learn AI?

Definition:

Learning AI helps us understand and build intelligent systems that can automate tasks and solve real-world problems.

Explanation:

AI is widely used in industries like healthcare, banking, and IT. It improves efficiency and provides better career opportunities.

3. What is an AI agent?

Definition:

An AI agent is an entity that perceives its environment through sensors and acts upon it using actuators to achieve a goal.

Explanation:

It observes → thinks → acts.

👉 Example: A robot vacuum cleaner.

4. Examples of AI agents

Definition:

AI agents are systems that can sense and act intelligently in an environment.

Examples:

- Virtual assistants (Siri, Alexa)

- Self-driving cars
 - Chatbots
 - Game-playing AI
-

5. Advantages of AI (any 4)

Definition:

Advantages of AI are the benefits gained by using intelligent systems.

Points:

- Reduces human effort
 - High accuracy
 - Works 24/7
 - Fast decision-making
 - Handles risky tasks
-

6. What is Uninformed Search?

Definition:

Uninformed search is a search strategy that explores possible solutions without using any additional knowledge about the goal.

Explanation:

It blindly searches until it finds a solution.

👉 Examples: BFS, DFS.

7. What is AI environment?

Definition:

An AI environment is the surroundings in which an AI agent operates and interacts.

Explanation:

It includes all external factors affecting the agent.

👉 Example: Roads and traffic for a self-driving car.

5 Marks Answers

1) Types of AI Environment

1. Fully Observable vs Partially Observable

Fully Observable Environment

In a **fully observable environment**, the AI agent can perceive the **complete state of the environment at all times**. All the necessary information required to make a decision is available to the agent. Because the agent has full knowledge, decision-making becomes easier and more accurate.

Example:

In a **chess game**, both players can see the entire board and all pieces, so the environment is fully observable.

Partially Observable Environment

In a **partially observable environment**, the agent can only observe **limited or incomplete information** about the environment. Some important details may be hidden or uncertain. Therefore, the agent must make decisions using **incomplete data or predictions**.

Example:

A **self-driving car** cannot see everything around it due to obstacles, weather conditions, or sensor limitations.

2. Deterministic vs Stochastic

Deterministic Environment

In a **deterministic environment**, the next state of the environment is **completely determined by the current state and the agent's action**. This means there is no randomness or uncertainty involved.

Example:

Solving a **mathematical problem** or a puzzle where the same input always produces the same result.

Stochastic Environment

In a **stochastic environment**, the next state is **uncertain or influenced by probability**. Even if the same action is performed, the result may differ each time due to randomness or external factors.

Example:

Weather prediction, where conditions may change unpredictably.

3. Episodic vs Sequential

Episodic Environment

In an **episodic environment**, each task or decision is **independent of previous actions**. The agent performs an action based only on the current situation without considering past decisions.

Example:

In **image recognition**, each image is analyzed separately and does not affect future analyses.

Sequential Environment

In a **sequential environment**, the agent's current action **affects future states and decisions**. Therefore, the agent must consider past actions and plan carefully.

Example:

In **chess**, every move affects the future moves and the outcome of the game.

4. Static vs Dynamic

Static Environment

A **static environment** does not change while the agent is making decisions. The agent has enough time to analyze the environment before taking action.

Example:

A **crossword puzzle**, where the puzzle remains the same while solving it.

Dynamic Environment

A **dynamic environment** changes continuously, even while the agent is making decisions. The agent must react quickly to these changes.

Example:

Driving a car, where traffic, pedestrians, and signals keep changing.

5. Discrete vs Continuous

Discrete Environment

In a **discrete environment**, the number of states, actions, and possible outcomes are **limited and countable**. These environments usually involve clear and defined steps.

Example:

Board games like **chess** or **tic-tac-toe**.

Continuous Environment

In a **continuous environment**, the number of states and actions is **infinite and constantly changing**. The agent must deal with continuous values such as time, speed, or distance.

Example:

Robot movement or controlling the speed of a self-driving car.

6. Single-Agent vs Multi-Agent

Single-Agent Environment

In a **single-agent environment**, only one intelligent agent operates and makes decisions independently.

Example:

A computer solving a **puzzle** or **Sudoku**.

Multi-Agent Environment

In a **multi-agent environment**, multiple agents interact with each other. These agents may **cooperate or compete** while achieving their goals.

Example:

A **traffic system**, where multiple vehicles interact with each other.

2. explain the following ai agent

1. simple reflex
2. model base

1. Simple Reflex Agent

Definition:

A **Simple Reflex Agent** is an AI agent that makes decisions based only on the current percept (current input from the environment). It follows predefined condition–action rules (if–then rules) to perform actions. This agent does not store past information or memory. It works well in fully observable environments.

Example: Thermostat that turns the heater on when temperature is low.

Key Characteristics

- Works using **condition–action rules** (if–then rules).
- Does **not store previous information**.
- Suitable for **fully observable environments**.
- Simple and fast but limited intelligence.

Working Principle

- The agent observes the environment.
- Matches the current situation with predefined rules.
- Executes the corresponding action.

Typical Use Cases

- Basic automation systems
- Simple game agents
- Temperature control systems

Advantages

- Easy to design
- Quick response

Limitations

- Cannot handle partially observable environments
 - No memory of past events
-

2. Model-Based Agent

Definition:

A **Model-Based Agent** is an AI agent that maintains an internal model or memory of the environment. It uses both the current percept and past information to decide the best action. This helps the agent work in partially observable environments. The internal model updates as the environment changes.

Example: Robot vacuum cleaner remembering obstacles in a room.

Key Characteristics

- Maintains an **internal state (memory)**.
- Uses a **model of how the world works**.
- Can work in **partially observable environments**.

Working Principle

- Observes the current percept.
- Updates its **internal model** based on past experiences.
- Chooses an action using rules and stored knowledge.

Typical Use Cases

- Autonomous robots
- Self-driving vehicles
- Smart home automation

Advantages

- Handles incomplete information
- More intelligent than simple reflex agents

Limitations

- More complex to design
- Requires memory and processing power

3) Applications of AI Agents

1. Robotics

AI agents are used in robots to help them **sense the environment and perform tasks automatically**. They collect information through sensors and make decisions to complete work efficiently. These robots are widely used in industries, hospitals, and homes. AI agents improve **speed, accuracy, and productivity** in robotic systems.

Example: Industrial robots used in factories for assembling products.

2. Autonomous Vehicles

AI agents are used in **self-driving cars and smart transportation systems**. They analyze data from cameras, sensors, and GPS to detect obstacles, follow traffic rules, and choose safe routes. This helps vehicles drive safely without constant human control. AI agents also help reduce accidents and improve traffic management.

Example: Self-driving cars that automatically navigate roads.

3. Virtual Assistants

AI agents power **virtual assistants and chatbots** that communicate with users using voice or text. They understand user commands, answer questions, and perform tasks such as setting reminders or searching for information. These assistants improve user convenience and productivity. They are commonly used in smartphones and smart devices.

Example: Smartphone voice assistants that answer questions and manage daily tasks.

4. Gaming

In video games, AI agents control **non-player characters (NPCs)** and make them behave intelligently. They react to player actions, plan strategies, and create a more realistic gaming environment. AI agents make games more interactive and challenging. This improves the overall gaming experience for players.

Example: Enemy characters in games that react to the player's movements.

5. Healthcare

AI agents help doctors and healthcare professionals **analyze medical data and diagnose diseases**. They can examine medical images, predict health conditions, and suggest treatments. This improves the accuracy and speed of diagnosis. AI agents also assist in patient monitoring and medical research.

Example: AI systems used to detect diseases from medical images like X-rays.

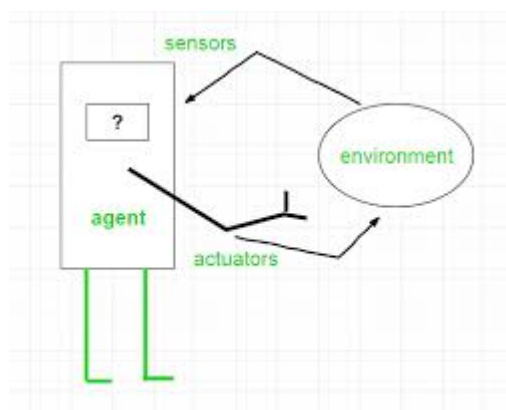
6. E-Commerce and Recommendation Systems 🛒

AI agents analyze **customer behavior, search history, and preferences** to recommend products or services. They help online platforms provide personalized suggestions to users. This improves customer satisfaction and increases business sales. AI agents are widely used in online shopping and streaming platforms.

Example: Online shopping websites recommending products based on user searches.

4) Structure of an AI Agent 🤖

The **structure of an AI agent** describes how an intelligent system is organized to **perceive its environment, process information, and take actions to achieve a goal**. An AI agent interacts with its surroundings using **sensors and actuators**, while the internal program decides the appropriate action. The structure helps the agent behave intelligently and respond to changes in the environment.



1. Sensors 👁️

Sensors are the components that allow the AI agent to **observe and collect information from the environment**. They detect different types of input such as images, sound, temperature, or movement. The information received from sensors is called **percepts**. Sensors help the agent understand the current state of the environment before making decisions.

Examples: Cameras in robots, microphones in voice assistants, temperature sensors in smart devices.

2. Agent Program 🧠

The **agent program** is the main decision-making part of the AI agent. It is a software program that **processes the percepts received from sensors and decides what action to perform**. The program may use rules, algorithms, or machine learning techniques to determine the best action. It runs on a physical system called the **agent architecture**.

Example: A navigation program in a self-driving car that decides when to stop or turn.

3. Actuators ⚙️

Actuators are the components that allow the AI agent to **perform actions and affect the environment**. They convert the decisions made by the agent program into physical movements or outputs. Actuators help the agent interact with the outside world.

Examples: Robot arms, wheels, speakers, display screens, or motors.

4. Environment 🌍

The **environment** is the surroundings in which the AI agent operates. It includes all objects, conditions, and other agents that the system interacts with. The environment provides inputs to the agent and is affected by the agent's actions. AI agents must continuously observe and adapt to changes in the environment.

Examples: Road conditions for a self-driving car or a home environment for a cleaning robot.

5. Agent Architecture 💻

Agent architecture refers to the **hardware or platform on which the agent program runs**. It provides the necessary computing power and resources for the agent to function. The combination of **agent architecture and agent program forms a complete AI agent system**.

Examples: Computers, robots, smartphones, or embedded systems.